

Review

Electrocardiographic changes associated with epilepsy beyond heart rate and their utilization in future seizure detection and forecasting methods

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HIGHLIGHTS

- People with epilepsy (PWE) show peri-ictal ECG abnormalities outside of heart rate.
- Cardiac conduction abnormalities such as in QTc and PR intervals are common in PWE.
- Using ECG changes outside of heart rate may help develop seizure forecasting methods.

ABSTRACT

The ability to assess seizure risk may help provide timely warnings and more personalized treatment plans for people with epilepsy (PWE). ECG changes are commonly observed in epilepsy which make ECG a promising candidate to monitor seizure risk. Most ECG research in this domain has focused on heart rate-related changes. However, several studies have identified a range of other peri-ictal ECG parameter changes that may potentially prove useful for seizure detection and forecasting. Here, we offer a systematic review of ECG changes in epilepsy outside of heart rate. We performed the systematic literature review according to PRISMA guidelines using key words related to ECG, SUDEP and epilepsy. We identified and screened 502 abstracts, read 110 full papers, and included 24 papers in the final review. Our results suggest that PWE may be more prone to cardiac conduction abnormalities than healthy controls. During interictal periods, PWE were more likely to have abnormal QTc intervals, ST segment abnormalities, elevated T Waves, early repolarization (ER), increased P Wave dispersion and PR intervals when compared to controls. Apart from these baseline abnormalities, changes during the pre-ictal and ictal states have been reported, with arrhythmias, QTc prolongation and ST segment changes being the most common. A better understanding of these state-dependent changes may afford less-cumbersome and less-stigmatizing epilepsy monitoring tools in the future.

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1. Introduction

This review provides a survey on epilepsy associated electrocardiographic (ECG) changes outside of heart rate, some of which may be useful in developing seizure detection and forecasting methods. Reliable methods to detect and forecast epileptic seizures may help alleviate a major burden for epilepsy patients, (Landover, 2016) and support clinicians in delivering more objective, personalized treatments (Meisel and Loddenkemper, 2019). Fueled by long-term data and novel analysis approaches, recent years have seen a surge of such increasingly accurate methods to monitor, detect, and predict seizures in epilepsy (Mormann and Andrzejak, 2016; Kuhlmann et al., 2018). However, in order to ensure that seizure detection and forecasting are not just feasible, but implemented into day to day epilepsy care, easy to apply and monitor modalities amenable to everyday use are crucial (Dumanis et al., 2017). Assessing seizure risk has therefore for several decades also considered electrocardiography (ECG). In contrast to EEG and ECoG, ECG can be measured with minimal complexity, non-invasively and with less stigma, which facilitates implementation in broad

ambulatory use. The overwhelming amount of research using ECG for such a purpose has thereby focused on heart rate and heart rate-related variables (Bruno et al., 2018). Motivated by research showing that heart rate increases at seizure onset (Eggleston et al., 2014), increases in heart rate are, for example, currently used to trigger responsive Vagus nerve stimulation aimed at preventing or shortening epileptic seizures (Boon et al., 2015; Fisher et al., 2016). Apart from seizure detection, changes in heart rate and heart rate variability have also been tested in seizure forecasting research (Kerem and Geva, 2005; Valderrama et al., 2012; Fujiwara et al., 2016; Pavei et al., 2017).

While heart rate and heart rate-related ECG variables have long been the focus of epilepsy research, producing many excellent reviews (Bruno et al., 2018), other elements of the ECG and their potential role for seizure detection and forecasting have remained less explored. In a recent study, it was found that single-channel ECG contained a comparable amount of information about the pre-ictal state to scalp EEG, and in this study whole frequency spectrum ECG assessment contributed to the pre-ictal state evaluation, not solely heart rate and related variables. When only heart

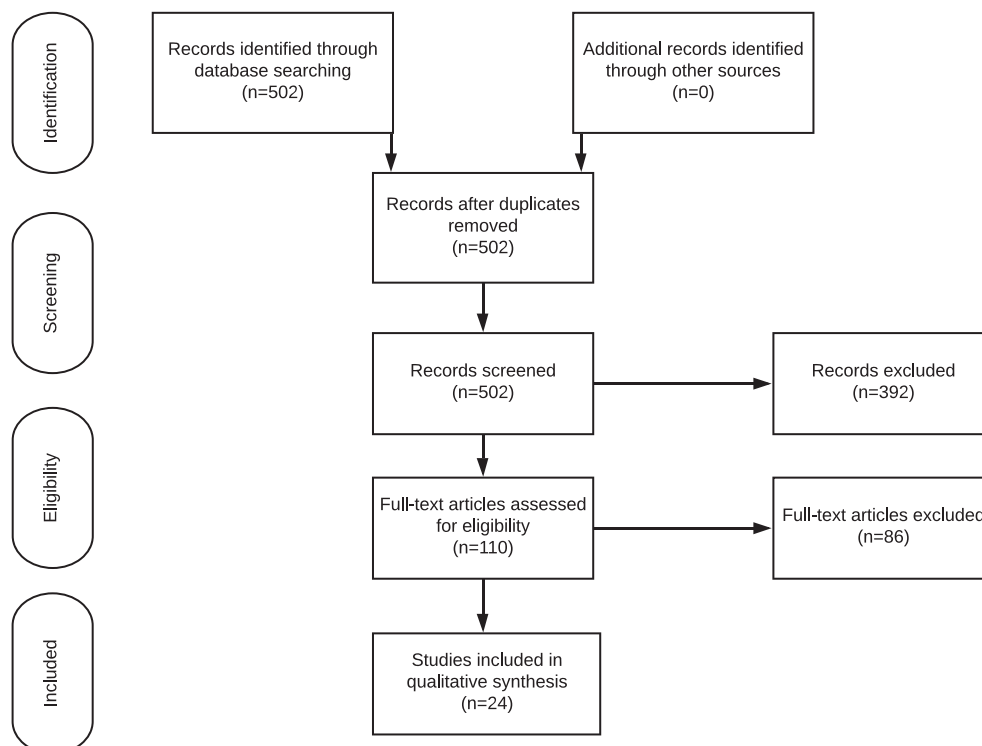


Fig. 1. Search strategy according to the Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA) guidelines.

Table 1
Summary of article findings identified by the PRISMA search.

Study	Number of patients	Age	Gender	Number of seizures	ECG characteristics investigated	Seizure or epilepsy types, etiology	Pre-ictal definition	Ictal definition	Post-ictal definition	Interictal definition	Results
(Moseley and Britton, 2014) (Cohort Study)	29	Not stated	Not stated	58	– QTc changes	– Focaldyscognitive without secondary generalization – Primarily and secondarily generalized tonic-clonic seizures	2 min prior to seizure	15 s, 30 s, 45 s, 60 s, 90 s, 120 s after seizure onset	2 min after seizure offset	N/A	– Ictal QTc lengthening occurred more than shortening
(Özdemir et al., 2016) (Cohort Study)	70	59 ± 39 months	32 female	Not stated	– P wave dispersion – QT/QTc interval and dispersion – T _{peak} –T _{end}	– Diverse population including drug resistant epilepsy, hypoxic-ischemic encephalopathy, West Syndrome and more – DS and other epilepsy syndromes	N/A	N/A	N/A	90 s	– Under ketogenic diet all ECG parameters except P wave dispersion increased (not significantly) after 12 months
Delogu et al. (2011) (Cohort Study)	– 20 patients with Dravet Syndrome (DS) – 20 patients with other epilepsy syndromes on ASM – 20 patients with other epilepsy syndromes not on ASM – 20 healthy controls	– DS Group: 6.8 ± years – ES/ ASM: 6.0 ± 4 years – ES/no-ASM: 6.7 ± 4 years – HC: 7.2 ± 5 years	– DS: 11 female – ES/ ASM: 12 female – ES/no-AED: 10 female – HC: 11 female	193	– QT/QTc intervals	– DS and other epilepsy syndromes	N/A	N/A	N/A	24 hours	– QT/QTc intervals not different between DS and control groups
Brotherstone et al. (2010) (Cohort Study)	39	17.6 years	14 female	156	– QT intervals	– Absence seizures – Generalized tonic-clonic seizures – Temporal, frontal lobe seizures – Subclinical seizures	5 min prior to seizure	Start of seizure to end	N/A	N/A	– Statistically significant increase in QTc occurred ictally in some patients
Opherk et al. (2002) (Cohort Study)	41	35 ± 13.3 years	22 female	102	– Sinus tachycardia – ST depression – T wave inversion	– Complex partial-Primarily and secondarily generalized	N/A	Start of seizure to end	1–2 min post-ictal	N/A	– Potentially serious ECG abnormalities, including ST depression and T wave inversion are not unusual, especially in in generalized seizures (observed in 35%)
Verrier et al. (2016) (Cohort Study)	28	Mean 40 years	19 female	Not Stated	– T wave alternans – Fluctuations of ST segment or T wave	– Drug-resistant epilepsy dominated by focal seizures	N/A	N/A	N/A	Yes	– T wave alternans severely elevated interictally, but improvement through vagus nerve stimulation
Tigaran et al. (2003) (Cohort Study)	23	Not stated	10 female	125	– ST segment depression	– Drug-resistant epilepsy with simple and complex partial seizures	4 min prior to seizure	Start of seizure to end	Not stated	N/A	– ST segment changes prior to EEG seizure activity

Table 1 (continued)

Study	Number of patients	Age	Gender	Number of seizures	ECG characteristics investigated	Seizure or epilepsy types, etiology	Pre-ictal definition	Ictal definition	Post-ictal definition	Interictal definition	Results
						without secondary generalization as well as secondarily generalized tonic-clonic seizures					– ST segment changes ictally and associated with generalized seizures – In 36% of patients ST segment change occurred immediately post-ictally. – ER and severe QTc prolongation appear more prevalent in people in epilepsy
Lamberts et al. (2015)a (Cohort Study)	185 people with epilepsy, 178 controls	– Epilepsy cohort: 38 ± 13.3 years – Control cohort: 48 ± 12.5 years	– Epilepsy cohort: 100 female – Control cohort: 113 female	0	– QTc interval – Brugada ECG pattern – ER	– Drug-resistant epilepsy	N/A	N/A	N/A	Yes	
Lamberts et al. (2015)b (Case-control Study)	18 people with epilepsy and ventricular tachycardia (VT) or ventricular fibrillation (VF), 470 controls with VT/VF without epilepsy, 54 individuals with epilepsy without VT/VF	– Cases: 57 ± 16.9 years – VT/VF controls: 64 ± 14.0 years – Epilepsy controls: 57.3 ± 15.9 years	– Cases: 6 female – VT/VF controls: 94 female – Epilepsy controls: 18 female	Not Stated	– VT/VF	– Active epilepsy	N/A	N/A	N/A	Yes	– Cardiovascular disease was predominant cause of VT/VF in epilepsy rather than epilepsy characteristics
(Krishnan and Krishnamurthy, 2013) (Cohort Study)	40 with non-epileptic seizures (NESs), 39 with probable epilepsy (PE), 116 with definite epilepsy (DE)	– NESs: 38.7 ± 1.2 years – PE: 44.8 ± 2.2 years – DE: 43.2 ± 1.2 years	– NESs: 28 female – PE: 24 female – DE: 66 female	0	– PR interval – Mean QRS electrical axis	– Non-epileptic seizures, probable epilepsy (focal, left or right and generalized onset), definite epilepsy	N/A	N/A	N/A	Yes	– Patients with localized epilepsy had significantly lower PR interval compared to patients with non-epileptic seizures – Mean QTc not significantly different between the three groups – Antiepileptic polytherapy associated with significantly lower QTc interval
(Sadrnia et al., 2013) (Cohort Study)	254 people with epilepsy, 254 controls	Not Stated	– PWE: 110 female – Controls:	0	– QTc interval	– Primary seizures with unknown etiology	N/A	N/A	N/A	Yes	– Children with unknown seizure causes have significantly longer

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Table 1 (continued)

Study	Number of patients	Age	Gender	Number of seizures	ECG characteristics investigated	Seizure or epilepsy types, etiology	Pre-ictal definition	Ictal definition	Post-ictal definition	Interictal definition	Results
											QTc intervals
Ergul et al. (2013) (Cohort Study)	15 patients with DS, 20 healthy controls	– DS: 8.2 ± 4 years – HC: 8.1 ± 3 years	– 109 female – DS Group: 6 female – HC Group: 7 female	0	– QRS axis – Repolarization abnormalities – QTc dispersion – P wave dispersion – PR interval	– DS	N/A	N/A	N/A	Yes	– P wave dispersion, QT and QTc dispersion significantly higher in DS – No abnormality in QRS axis, PR intervals and ST and T waves
(Hayashi et al., 2019) (Cohort Study)	354 people with epilepsy, 140 age- and sex-matched controls	– PWE: 33 ± 16 years – Controls: 31 ± 12 years	– PWE: 162 female – Controls: 70 female	0	– J peak and ST segment	– Generalized and focal epilepsy	N/A	N/A	N/A	Yes	– Overall incidence of abnormal ER with J peak more than or equal to 0.1 mV was higher in the epilepsy group – ER with horizontal or descending ST segment more prevalent in epilepsy patients – SUDEP occurred in two patients, one with abnormal ER in 7 year follow up
Kolsal et al. (2014) (Cohort Study)	20 patients with refractory epilepsy, 20 patients with well-controlled epilepsy, 20 healthy controls	– Group 1: 9.55 ± 5.02 years – Group 2: 10.1 ± 4.18 years – Group 3: 10.35 ± 4.39 years	– Group 1: 8 female – Group 2: 8 female – Group 3: 8 female	17	– QT/QTc intervals and dispersion – PR interval	– Drug-refractory epilepsy and well-controlled epilepsy	15 min prior to seizure	Start of seizure to end	30 min after seizure	N/A	– One patient in drug-refractory group had T wave inversion during seizure – More patients with significant QTc lengthening interictally in drug-refractory group compared to controlled epilepsy group
Fialho et al. (2017) (Cohort Study)	30 people with epilepsy, 30 age- and sex-matched controls	– PWE: 37.4 ± 11.2 years – Controls: 37.4 ± 11.2 years	– PWE: 12 female – Controls: 12 female	0	– Chronotropic incompetence	– Temporal lobe epilepsy	N/A	N/A	N/A	Yes	– Chronotropic incompetence more common in patients with epilepsy

Table 1 (continued)

Study	Number of patients	Age	Gender	Number of seizures	ECG characteristics investigated	Seizure or epilepsy types, etiology	Pre-ictal definition	Ictal definition	Post-ictal definition	Interictal definition	Results
De Sousa et al. (2017) (Cohort Study)	62 people with epilepsy, 57 age- and sex-matched controls	PWE: 39.58 ± 14.6 years	PWE: 35 female	0	- Cardiac electric axis - P wave duration - PR interval - QT/QTc intervals and dispersion	Symptomatic focal epilepsy	N/A	N/A	N/A	Yes	- No ST segment deviations for both groups - PWE had significantly prolonged P wave, PR interval, QTc interval, pathologic QT dispersion, and left atrial overload - ER significantly higher in patients with epilepsy
(Brotherstone and Mclellan, 2012) (Cohort Study)	11	3–60 years	4 female	33	Subclinical seizures	- Cardiac index of parasympathetic activity (CIPA) - Vagal tone by measuring RR interval and QRS signal	1 min prior to seizure	Start of seizure to end	Not Stated	Segments recorded from stages 3 or 4 of sleep	- CIPA decreased from baseline during seizures - Generalized seizures were associated with increase in CIPA
Akalin et al. (2003) (Cohort Study)	28 people with epilepsy and no ASM, 34 people with epilepsy on ASM, 52 controls	- Group 1: 6.8 ± 3.92 years - Group 2: 7.51 ± 3.68 years - Group 3: 6.94 ± 3.92 years	- Group 1: 13 female - Group 2: 16 female - Group 3: 25 female	0	Newly diagnosed epilepsy with idiopathic or cryptogenic etiology	- QT/QTc intervals and dispersion - P wave amplitude and duration - PR interval - QRS duration and axis	N/A	N/A	N/A	Yes	- QT dispersion increased in children with epilepsy - No difference in P wave amplitude or duration, PR interval, QRS duration, QRS axis, QT or QTc intervals - Antiepileptic drug treatment did not affect QT dispersion
Surges et al. (2010b) (Cohort Study)	25	34 ± 1.6 years	Not Stated	50	- Drug-refractory temporal lobe epilepsy - Complex partial seizures - Secondarily generalized seizures	- QTc intervals - Cardiac rhythm	N/A	Start of seizure to end	1, 3 and 5 min after seizure	Not Stated	- Abnormal ictal QTc prolongation occurred in three patients, no association with GTCs - Benign cardiac arrhythmias in 14 patients independent of seizure type - Abnormal QTc shortening in 17 patients during post-ictal phase and more often in secondarily GTCs
(Stefani et al., 2013) (Cohort Study)	36	41 ± 11.5 years	21 female	Not Stated	Drug-refractory epilepsy	QT, QTc intervals	N/A	N/A	N/A	10-minute epochs	QT interval shortened when off ASM

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Table 1 (continued)

Study	Number of patients	Age	Gender	Number of seizures	ECG characteristics investigated	Seizure or epilepsy types, etiology	Pre-ictal definition	Ictal definition	Post-ictal definition	Interictal definition	Results
(Rugg-Gunn et al., 2004) (Cohort Study)	20	40 ± 16 years	10 female	3377	– Drug-refractory simple partial seizures – Complex partial seizures – Secondarily generalized seizures	– Asystole	8 min prior to seizure	Start of seizure to end	2 min after seizure	Yes	– No patients had a pathologically long QT interval and no significant changes during sleep – Ictal bradycardia (<40bpm) was rare, four patients had periods of asystole that needed permanent pacemaker
Garson et al. (1993) (Cross-Sectional Study)	287	6.8 ± 5.6 years	141 female	0	– Patients with LQTS and/or seizures	– Long QT Syndrome (LQTS)	N/A	N/A	N/A	Yes	– Of patients with LQTS, 10% had seizures
Schimpf et al. (2012) (Cohort Study)	19	41 ± 12 years	7 female	0	– Lennox Gastaut syndrome – Epilepsy with focal or generalized seizures	– QRS complex – QT interval – T_{peak}-T_{end} – Syncope	N/A	N/A	N/A	Yes	– Mean interictal QTc interval shortened significantly following rufinamide administration – T_{peak}-T_{end} significantly shorter in patients following rufinamide administration
Surges et al. (2010a) (Cohort Study)	19 living controls with epilepsy, 19 SUDEP controls	– Living: 36.2 ± 9.1 years – SUDEP: 36.1 ± 9.1 years	– Living: 8 female – SUDEP: 8 female	89	– Drug-refractory focal epilepsy	– QTc interval – QT dispersion	Not Stated	Start of seizure to end	5 minutes after seizure	Yes	– Patients with refractory epilepsy have abnormal cardiac repolarization at rest, during or after seizures – QTc prolongation above pathologic upper limits occurred in 10.11% of seizures and 13.16% of patients, 26.4% of patients had pathologic QT dispersion. – 44.7% patients had abnormal QTc shortening post-ictally

Abbreviations: ASM: Antiseizure Medication/s, ER: Early repolarization, DS: Dravet Syndrome.

rate-based measures were used, seizure forecasting algorithms performed far worse (Meisel and Bailey, 2019). These observations raise questions regarding details of other ECG features beyond heart rate occurring peri-ictally. A better understanding of these ECG features may not only help devise improved seizure detection and forecasting algorithms, but may also shed new light on mechanisms underlying ictal autonomic changes. For these reasons, we performed a systematic review of the literature-reported peri-ictal ECG changes outside of heart rate.

2. Methods

We conducted a systematic review according to Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA) guidelines (Moher et al., 2009) (Fig. 1). Inclusion criteria consisted of studies (1) written in English, (2) reporting randomized controlled trials, cohort studies or cross-sectional studies, or, more generally, reporting findings from original research studies (thus excluding editorials, reviews, etc.) (Fig. 1), (3) including a report of at least one person with epilepsy (PWE) who was alive at the time of the study, (4) mentioning ECG, and (5) reporting on peri-ictal ECG changes. Consequently, we excluded papers not in English, reviews, letters, editorials, conference proceedings, case reports, postmortem and animal studies, and unpublished work. We performed the literature search in PubMed, Cochrane, Embase and Cinahl, including all articles published from 1993 until January 2019. CU, RE and a librarian CR, developed the search strategy. We used a combination of keywords: SUDEP, epilepsy, seizure, ECG, and various ECG features such as P waves, QRS complex, and QTc interval. Example MeSH term keywords included “sudden unexpected death in epilepsy,” “sudden death in epilepsy,” and “SUDEP.” Similarly, we used “epilepsy,” “epilepsies,” and “epileptic,” among other search terms. Papers included for potential review were managed on an internal reference library and an electronic data entry tool. Our initial search strategy yielded 502 papers. After reviewing titles and abstracts, 110 papers remained. Following a review of these full texts, 24 papers fit the inclusion criteria. To select papers, two independent reviewers (CU and RE) had to agree on the inclusion of each paper. If there was disagreement, a third reviewer (CM) decided on the inclusion or exclusion of the paper. Data variables included age, gender, number of study participants, ictal state analyzed use of anti-seizure or cardiac medications, and seizure type. We classified data that related to the ECG findings for the peri-ictal states.

3. Results

The main goal of our work was to provide a systematic review of ECG changes in epilepsy beyond heart rate-related changes. Our systematic review according to the Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA, Fig. 1) guidelines yielded 24 articles to be included for final review (Table 1) (Moher et al., 2009). A particular focus of our work was on changes occurring prior to or during seizures which could potentially be used as warning or detection signs, respectively, but also on any additional changes occurring peri-ictally. We thus report changes in four categories, interictal, pre-ictal, ictal, and post-ictal. Our interictal group included 19 studies, the pre-ictal group contained 1 study, the ictal group contained 9 studies, and post-ictal changes were reported in 3 studies. Table 1 provides a summary of the main findings reported in the articles. In this table, we list the time definitions for the respective states for better comparison across studies and to account for the temporal change of ECG alterations with respect to seizure onset.

3.1. Pre-ictal ECG changes (one study)

3.1.1. ST segment changes

Pre-ictal ECG changes were overall only rarely the focus of the studies identified by our search. One study reported pre-ictal, defined as 4 minutes prior to seizure, ST segment depression (Tigaran et al., 2003). Specifically, Tigaran et al. compared 22 patients with drug-refractory epilepsy to healthy patients. The authors observed that ST segment changes were not seen in the healthy patients control group, even with maximal exertion (Tigaran et al., 2003).

3.2. Ictal ECG changes (nine studies)

3.2.1. QTc interval changes

QT abnormalities were by far the most commonly reported ECG feature. Corrected QT (QTc) interval changes, predominantly QTc prolongations, were reported frequently during ictal ECG recordings. Notably, many studies determined individual cut-offs for marked QTc prolongation and QTc prolongation severity. Moseley et al. reported significant ictal QTc lengthening in 5.2% of patients and in 19% of seizures ($n = 29$) (Moseley and Britton, 2014). Brotherton et al. reported all of their 39 patients to exhibit significantly longer ictal QTc intervals, compared to the pre-ictal period (5 minutes prior to seizure onset), even if the overall interval was still within normal limits (Brotherton et al., 2010). Twenty-three percent of patients experienced QTc prolongation (>512 ms). Surges et al. reported abnormal QTc prolongation during the ictal and post-ictal phases in 12% of their subjects (Surges et al., 2010b). They also found benign arrhythmias in 56% of patients, independent of seizure type. In another study, Surges et al. looked at ictal cardiac features in 91 seizures from 19 PWE and 19 deceased SUDEP patients (Surges et al., 2010a). They found seizure-related QTc prolongation of more than 10 ms in 48.3% of seizures and in 63.16% of patients. Twenty-five seizures and 24 patients with significant QTc prolongation were from patients who died of SUDEP. QTc prolongation occurred in upper limits in 10.1% of seizures and 13.2% of patients in this group. This study also reported QTc values below normal limits in one PWE and two SUDEP patients. Lateralization of seizure onset did not affect QTc prolongation (Surges et al., 2010a). Similarly, Surges et al. also found 9 out of their 38 patients had pathologic QTc dispersion (>50 ms) of which 5 were SUDEP patients (Surges et al., 2010a).

3.2.2. ST segment changes/T wave inversion

ST segment abnormalities were less frequently reported but appeared more severe when they did occur. Opherke et al. compared interictal to ictal cardiac rhythm and conduction abnormalities in 102 seizures from 41 patients (Opherke et al., 2002). Out of 21.5% of seizures that showed ECG abnormalities, 6% could potentially be serious, or not benign, and were not observed pre-ictally. Potentially serious abnormalities included ST depression occurring in five seizures from one of the 41 patients, and T wave inversion occurring in one seizure (Opherke et al., 2002).

Overall, ECG abnormalities most commonly occurred during generalized seizures, where they were prevalent in 35% of generalized seizures. Tigaran et al. found that 40% of 22 patients with drug-refractory epilepsy experienced ST segment depression at the offset of their seizures (Tigaran et al., 2003). In this study, 77% of seizures associated with ST segment depression occurred during sleep. ST segment changes were also more frequently associated with generalized seizures. The authors reported that ST segment changes were not common in patients without epilepsy, even with maximal exertion during a treadmill test. They also investigated comorbidities and found that duration of the seizure, sex,

and seizure onset location did not predict ST segment change (Tigaran et al., 2003).

Kolsal et al. compared ictal ECG characteristics in 20 patients with drug refractory epilepsy to 20 patients with drug-controlled epilepsy and to baseline ECGs in 20 healthy patients (Kolsal et al., 2014). Of all the patients analyzed, one patient in the drug refractory epilepsy group had T wave inversion during ECG monitoring. This abnormality occurred during 3 focal and 1 generalized seizure (Kolsal et al., 2014).

3.2.3. QRS axis

Two studies looked at QRS axes changes in the ictal state. Brotherstone et al. measured autonomic instability in subclinical seizures by comparing ictal to interictal periods from 33 seizures in 11 PWE (Brotherstone and McLellan, 2012). They analyzed the cardiac index of parasympathetic activity (CIPA), which assesses the vagal tone by comparing the incoming QRS signal to a previously acquired QRS template (Brotherstone and McLellan, 2012). Specifically, CIPA measures vagal tone by observing “the increase in pulse interval per unit increase in systolic pressure.” A low CIPA score (1–4) represents low vagal tone. The authors found that during the ictal period, generalized onset subclinical seizures were associated with increased CIPA and localized seizures were associated with decreased CIPA compared to the interictal period. Decreased CIPA was taken as an indication of a decreased vagal tone and cardiac parasympathetic activity (Brotherstone and McLellan, 2012). Rugg-Gunn et al. measured the frequency of ictal arrhythmias over 18 months in 20 patients with refractory focal seizures. Twenty percent (four) of their patients had periods of asystole that required permanent pacemaker insertion. Of those four patients, three had potentially fatal asystole. Most asystole occurred toward the end of the seizure. One patient had supraventricular tachycardia. The authors reported that the side of seizure lateralization did not affect these results (Rugg-Gunn et al., 2004).

3.3. Interictal (19 studies)

3.3.1. QTc Interval

As with the ictal phase, QTc abnormalities were reported as very common interictally. Lamberts et al. reported that PWE had significantly longer QTc intervals (405 ms, $n = 185$) during seizure-free periods than patients without epilepsy (294 ms, $n = 178$) (Lamberts et al., 2015a). Severe QTc prolongation was found in 5% of PWE versus 0% of patients without epilepsy. Delogu et al. compared QTc intervals in 20 healthy controls, 20 patients with Dravet Syndrome (DS), and 20 patients with other epileptic syndromes (Delogu et al., 2011). They found that, during a baseline 24-hour interictal recording, QTc intervals were not different between the three groups (Delogu et al., 2011). Similarly, Krishnan et al. found that mean QTc intervals were not different between their three groups of 116 PWE, 40 patients with non-epileptic seizures (NESs), and 39 patients with probable epilepsy (Krishnan and Krishnamurthy, 2013). They also reported that mean QTc intervals had a positive correlation with age and being female (Krishnan and Krishnamurthy, 2013).

Sadria et al. compared interictal QTc intervals of 254 PWE to 254 healthy controls (Sadria et al., 2013). The authors found 34.4% of patients with seizures had a QTc interval in the normal range (440 ms or less) while 12.6% of cases were on the border of being classified as long QT (between 440 ms and 460 ms), and 13% had QT intervals above the definition of long QT (>460 ms). These findings were compared to controls, of which 35% had a QTc interval between 440 ms and 460 ms, and only 9.4% had a QT interval greater than 460 ms. Syncope and sudden death risks were not different between the two groups (Sadria et al., 2013). De Sousa et al. reported similar findings when looking at interictal

intervals, defined as having no seizure that day in 62 PWE and 57 healthy controls (de Sousa et al., 2017).

PWE had, on average, longer QTc intervals during seizure-free periods. One of their 62 patients exhibited pathologic QTc prolongation during the study (de Sousa et al., 2017). Ergul et al. reported that QT and QTc values were similar between 15 patients with DS and 20 healthy controls when analyzing 24-hour interictal ECG recordings (Ergul et al., 2013).

Several studies investigated antiseizure medication (ASM) action and diet on cardiac activity. Stefani et al. analyzed effects on sodium channel blocking ASM withdrawal on interictal cardiac function in 36 patients with medically refractory epilepsy using ten-minute epochs of awake, REM and non-REM sleep (Stefani et al., 2013). ASM were completely weaned in 44.4% of their patients. The authors found none of their patients had pathologically long QT intervals. QT intervals shortened during withdrawal of sodium channel blocking ASM during wake periods but not during sleep. No other changes in ECG parameters were found in association with ASM withdrawal (Stefani et al., 2013). Schimpf et al. analyzed the effect of rufinamide treatment on cardiac features in 19 patients with Lennox-Gastaut and other epilepsy syndromes. They found that QT intervals shortened following rufinamide administration. Specifically, $T_{peak}-T_{end}$ intervals, defined as the difference of the interval of the Q wave to the peak of the T wave and the interval of the Q wave to the end of the T wave, were shorter. (Schimpf et al., 2012).

Ozdemir et al. analyzed the effects of Keto Diet (KD) on seizure suppression and cardiac function in 70 patients with epilepsy, focusing at P wave duration and dispersion, QTc dispersion and $T_{peak}-T_{end}$ intervals (Özdemir et al., 2016). The authors reported that all parameters except for P wave dispersion increased after 12 months on the KD, although these increases were not statistically significant. Overall, KD had no strong effect on cardiac function and repolarization (Özdemir et al., 2016). Akalin et al. reported no difference between 28 patients with newly diagnosed epilepsy, 34 patients under antiepileptic drug treatment, and 52 healthy controls in terms of QTc intervals (Akalin et al., 2003).

Indication for an association with QT changes and epilepsy has also come from studies investigating cardiology patients primarily. Garson et al. performed a community study looking at low and high-risk patients with Long QT Syndrome (LQTS). Of 287 patients investigated, 10% had seizures (Garson et al., 1993).

Interictal QT dispersion abnormalities appeared to be a commonality in PWE throughout many studies reviewed. Ergul et al. reported that PWE ($n = 15$) had a higher QT or QTc dispersion than controls ($n = 20$) (Ergul et al., 2013). Kolsal et al. found that at baseline, 50% of patients in the drug refractory epilepsy group ($n = 20$) and 25% of patients with drug-controlled epilepsy ($n = 20$) had abnormal QTc dispersions, defined as greater than 50 ms (Kolsal et al., 2014). Statistically significant QTc dispersion lengthening was only observed in patients with drug-refractory epilepsy (Kolsal et al., 2014). Surges et al. reported pathologic QTc dispersion in about 90% of PWE ($n = 19$). Conversely, they also found QTc values below normal limits in three out of 89 seizures (Surges et al., 2010a). Akalin et al. also demonstrated that QTc dispersion values were significantly increased in PWE compared to healthy controls. ASM treatment did not seem to affect QT dispersion (Akalin et al., 2003).

3.3.2. ST segment changes/T wave abnormalities

Interictal ST segment changes and T wave inversion abnormalities were reported in PWE in several studies; generally, these changes appeared less severe interictally than during ictal periods. Verrier et al. found that 68% of 28 patients with drug-resistant epilepsy exhibited severe T wave alternans (≥ 60 microvolts) (Verrier et al., 2016). The authors reported markedly elevated T

waves in 82% of patients (≥ 47 microvolts). This abnormality was decreased during vagus nerve stimulation in all patients (Verrier et al., 2016).

Fialho et al. tested cardiac stiffness of 30 PWE and 30 matched, healthy controls using a stress test (Fialho et al., 2017). They found no ST segment deviations for both groups during such tests. However, the authors reported that chronotropic incompetence, i.e. the inability of the heart to increase its rate in response to activity, was more common in PWE. Female gender, age, age of epilepsy onset, number of secondarily generalized seizures, as well as ASM polytherapy, were found to be associated with lower cardiovascular fitness (Fialho et al., 2017). Similarly, Ergul et al. found no interictal abnormalities in ST segments or T waves in precordial leads between 15 patients with DS and 20 healthy controls (Ergul et al., 2013).

A number of studies demonstrated that early repolarization (ER) may be a commonality in PWE. Hayashi et al. found ER to be more prevalent in their epilepsy group (26%) composed of 354 PWE versus an age- and sex-matched control group (12.9%, $n = 140$) (Hayashi et al., 2019). ER with horizontal or descending ST segments was more common in the epilepsy group (19.5% of patients) compared to age- and gender-matched controls (6.4%). ER was furthermore more prevalent in inferior leads in the PWE group. ER with an ascending ST segment occurred with equal frequency. SUDEP occurred in two of the patients in the study, one of them exhibiting abnormal ER (Hayashi et al., 2019). The authors found that the incidence of abnormal ER with a J peak more than or equal to 0.1 mV was higher in the epilepsy group (26%) (Hayashi et al., 2019). Generally, abnormal ER was more prevalent in PWE (19.8%) compared to age- and sex-matched controls (8.6%). The introduction of ASM did not change this affect. Male gender was found to be the only significant predictor of these ST segment abnormalities (Hayashi et al., 2019). Lamberts et al. found that PWE had significantly higher rates of ER (34%, $n = 185$) compared to controls without epilepsy (13%, $n = 178$) (Lamberts et al., 2015a). They associated this with a 1.7-fold increased risk of sudden cardiac arrest (SCA). De Sousa et al. similarly reported that ER was more common in PWE when compared to age- and sex-matched controls, although these results were still within normal limits (de Sousa et al., 2017).

3.3.3. P wave dispersion/PR interval

Through our search, we encountered mixed findings about the prevalence of P wave abnormalities in PWE interictally. If P wave abnormalities were reported interictally, they appeared more benign. Ergul et al. found that patients with DS had a significantly higher P wave dispersion compared to healthy controls, but that the maximum duration of P waves was similar between the two groups interictally (Ergul et al., 2013). The authors also found that the minimum P wave duration was shorter in DS patients but that PR intervals were the same between groups (Ergul et al., 2013). De Sousa et al. also found that the P wave and PR interval were more prolonged in PWE, although these results, as several of their other ECG findings, were still within normal limits (de Sousa et al., 2017). The authors furthermore reported left atrial overload and atrioventricular (AV) blocks to be more common in PWE. Male gender, older age and polytherapy were determined as risk factors for these ECG abnormalities (de Sousa et al., 2017). Krishnan et al. found a correlation with gender, suggesting that women with epilepsy had a longer PR interval (Krishnan and Krishnamurthy, 2013). The authors also reported that mean PR increases with age (Krishnan and Krishnamurthy, 2013). Akalin et al. found no differences in P wave amplitude, duration, or PR interval between PWE ($n = 62$) and healthy controls ($n = 52$) (Akalin et al., 2003).

3.3.4. QRS axis/QRS interval

QRS abnormalities did not seem to be common in PWE interictally according to our search. Krishnan et al. observed that PWE had a more leftward leaning QRS axis but found no change in overall QRS interval duration (Krishnan and Krishnamurthy, 2013). The authors also compared these features according to seizure types and found no difference between generalized and focal seizures for QRS and PT intervals as well as T waves (Krishnan and Krishnamurthy, 2013). De Sousa et al. reported similar findings with no differences in QRS interval when analyzing interictal ECG of PWE ($n = 62$) and age- and sex-matched controls ($n = 57$) (de Sousa et al., 2017). Ergul et al. investigated ECG abnormalities in patients with DS ($n = 15$) and found no abnormalities in QRS axis and QRS intervals (Ergul et al., 2013). Similarly, Akalin et al. reported no differences in QRS duration or QRS axis when comparing PWE ($n = 62$) to controls ($n = 52$) (Ergul et al., 2013).

3.3.5. Arrhythmias

Rugg-Gunn et al. measured the frequency of arrhythmias in 20 patients with refractory focal seizures (Rugg-Gunn et al., 2004). Twenty-one percent of patients had periods of asystole and 75% of these patients had potentially fatal asystole. Garson et al. analyzed risk factors for 287 patients with LQTS, a feature seen in PWE. Beyond the finding that 10% of patients with LQTS had seizures, ventricular arrhythmias were seen in 16% of initial routine ECGs of patients with LQTS and atrioventricular blocks occurred in 5% of overall patients (Rugg-Gunn et al., 2004). Lamberts et al. investigated whether causes of SCA differed between 18 patients who died of SUDEP and had ventricular tachycardia (VT) or ventricular fibrillation (VF), 54 PWE without VT/VF and 470 patients with VT/VF without epilepsy (Lamberts et al., 2015b). They found that myocardial infarction was the most common cause of VT/VF in PWE.

For patients who died of SUDEP, VT/VF occurred at a younger age (57 vs 64 years), occurred closer to home (89% vs 58%), and was less likely to be witnessed (72% vs 89%) compared to living epilepsy controls. They also reported the prevalence of congenital and inherited heart diseases to be higher in patients who died from SUDEP than in living controls with epilepsy.

It should be noted that these two studies (Rugg-Gunn et al., 2004; Lamberts et al., 2015b) collectively report on different at-risk populations: patients with focal onset seizures with autonomic linked asystole or bradycardia, patients with intrinsic heart disease and patients where arrhythmias or asystole may have resulted from respiratory-linked events. Thus, any evidence of T wave variability or cardiac microfibrosis linked to seizures may help to better disentangle the cause and attribute the risk for arrhythmias.

3.4. Post-ictal (three studies)

3.4.1. QTc abnormalities

Post-ictal abnormalities as a whole were less often analyzed than other states. However, in line with other studies, QTc abnormalities also seemed prevalent post-ictally. Surges et al. reported abnormal QTc shortening to occur in 68% of PWE immediately post-ictally ($n = 25$) (Surges et al., 2010b). The authors found this change to occur significantly more often in secondarily generalized tonic-clonic seizures (Surges et al., 2010b). In another study, Surges et al. found QTc shortening five minutes after seizure cessation in two out of 19 SUDEP patients (Surges et al., 2010a). The authors reported cardiac arrhythmias in nine out of their 38 patients which included PWE and SUDEP patients. These arrhythmias were considered benign and included premature atrial beats (Surges et al., 2010a).

3.4.2. ST segment/T wave inversion

Only one study reported post-ictal ST segment changes. Specifically, it was found that of all the patients experiencing ST segment depression, 88% experienced it immediately after a seizure ($n = 23$ patients) (Tigaran et al., 2003).

4. Discussion

We provide an overview of peri-ictal ECG changes beyond heart rate. The effects of seizures on different subsystems of the autonomic nervous system (ANS), including cardiac function, have been shown for various seizure types (Mirsky and Van Buren, 1965; O'Regan and Brown, 2005). Assessing these cardiac function changes might provide insight into the pathophysiology of epileptic seizures and could help identify times when seizure risk is high. While heart rate and heart rate-related variables have long been the focus of extensive epilepsy research for these reasons (Bruno et al., 2018), other features in ECG might be equally or even more indicative of an impending or occurring seizure and other seizure complications, most important being SUDEP (Meisel and Bailey, 2019). More comprehensive assessments of ECG might help to develop better methods for automatic seizure detection or seizure forecasting. ECG based methods might help, to a certain extent, overcome some limitations of EEG or ECoG and be more applicable for everyday use in the ambulatory setting (Dumanis et al., 2017). Furthermore, a better understanding of the cardiac effects associated with seizures might help to gain important insights into the pathophysiology of seizures, insights that could potentially also help to better characterize patients at risk for SUDEP.

Following the PRISMA guidelines for systematic reviews (Moher et al., 2009) (Fig. 1) lead to the identification of 24 papers (Table 1) and demonstrated that the literature focused on these alternative ECG alterations is comparably limited. In particular the number of studies focusing on pre-ictal ECG changes was very low, despite the relevance that such changes might have in informing potential seizure forecasting methods. Often pre-ictal intervals were considered only for comparison to ictal states. A distinct comparison to how ECG conduction properties might already change during the pre-ictal state, however, was often not performed. Overall, ECG changes broadly related to cardiac repolarization (e.g. QT and ST intervals) appear to be the most reported or studied and were essentially reported for all states – pre-ictal, interictal, post-ictal and ictal – in the literature. In comparison to control groups, these changes also appear to be more prevalent in PWE.

The most common cardiac conduction abnormalities involved QTc prolongation and QTc dispersion. Most studies had different definitions for what was considered lengthening beyond normal limits, but studies concluded QTc interval prolongation to be a common abnormality in PWE. Several studies also reported the effects of antiseizure medications on QTc interval and found that they can also contribute to either lengthening or shortening. For instance, rufinamide and weaning off sodium channel blockers administration shortened QT intervals (Stefani et al., 2013; Schimpf et al., 2012) indicating that uncontrolled epileptogenicity, in the presence or absences of seizures, may independently contribute to autonomic changes. One study reported QTc shortening immediately after seizure offset; it is thus possible that seizures and the pre-ictal state might cause QTc prolongation and upon recovery, QTc shortening towards baseline. Similarly, several studies reported ST segment and T wave abnormalities. PWE were found to have a higher prevalence of these abnormalities and a higher risk of SCA as a result.

The pathophysiology behind these abnormalities and the relationship to epilepsy and seizures remains incompletely understood. It is possible that there may be underlying cardiac

conduction abnormalities in PWE (Moseley and Britton, 2014). In this case, stressful events such as seizures may serve as exacerbating circumstances, and therefore adding to cardiac distress. PWE may have abnormalities in repolarization. Because ST segments and QTc intervals are important markers of cardiac repolarization, this might be why there are often reported abnormalities (Lamberts et al., 2015a). For LQT- and asystole-associated events there is a significant risk that these events may actually be of primary cardiac origin instead of seizure, a concern particularly relevant when events are reported where EEG is not available. Furthermore, ST segment deviations, ictal or post-ictal, may likely constitute secondary effects induced by cardiac ischemia during seizures. The detection of such changes (LQT, asystole, ST segment deviations) should thus prompt a cardiac workup, including cardiac electrophysiological and coronary artery disease workup.

The frequent occurrence of ST and T wave changes during cardiac ischemia also highlights the importance of carefully considering the specificity and sensitivity of any of the reported ECG changes. As discussed, the ST segment deviation and T wave changes, for example, are likely secondary effects of ictal or post-ictal cardiac ischemia and thus not specific only to ictal events. Similarly, other conditions besides seizures are known to cause QTc prolongation. For example, exercise (Nie et al., 2019) and migraine (May et al., 2015) have been shown to be associated with QTc prolongation. The importance of carefully taking into consideration the sensitivity and specificity of electrophysiological changes (for example in EEG) has long been emphasized in epilepsy (Mormann et al., 2007). Thus, when considering any of the reported ECG changes, for example for seizure detection or forecasting methods, it will also be important to consider their specificity and sensitivity, similarly to approaches based on EEG or ECoG.

Seizures may arise or spread to areas of the brain that control autonomic function and hence cause autonomic alteration, such sympathetic or parasympathetic activation. These autonomic changes, which also play a role in SUDEP, may consequently manifest as changes in cardiac conduction (de Sousa et al., 2017; Ergul et al., 2013). Autonomic dysfunction furthermore contributes to other heart rate independent ECG changes including changes in QT intervals and ST segments (Stefani et al., 2013; Schimpf et al., 2012). Also, ketogenic diet might affect repolarization properties (Özdemir et al., 2016). While changes in heart rate, especially sinus tachycardia, seem to be the most common manifestation, other abnormalities do occur (Eggleston et al., 2014) and could ideally be picked up by future diagnostics based on ECG. Overall, while the causes underlying ECG changes may be multifaceted and should be carefully considered, the association of such ECG changes with epileptic seizures, in particular changes related to cardiac repolarization, suggests that ECG features beyond heart rate might be useful to consider for future seizure detection and forecasting methods.

Likewise, it is also important to note the different interactions that seizure type and etiology may have with the ECG. While Krishnan et al. (Krishnan and Krishnamurthy, 2013) observed no differences between focal and generalized seizures in terms of ECG abnormalities, other studies suggest generalized seizures are associated with greater ECG abnormalities than focal seizures. In particular, Tigaran et al. found a higher proportion of ST segment abnormalities in their generalized onset group (Tigaran et al., 2003). Similarly, studies by Surges and Fialho report evidence to suggest those with secondarily generalized seizures have lower overall cardiovascular fitness (Fialho et al., 2017; Surges et al., 2010b). As such, it is possible that generalized seizures are overall, more distressing to electrocardiac function. Table 1 provides a comprehensive summary of all reported changes along with the epileptic seizure types and more.

The effect of ASMs must also be considered as a possible effect on ECG parameter changes, in particular on QTc and PR intervals (Feldman and Gidal, 2013). Several studies included in our review specifically reported on the effects of ASM on QTc interval changes (Stefani et al., 2013; Akalin et al., 2003; Fialho et al., 2017). However, these studies also show that the overall consensus on ASM effect remains to be determined. Stefani et al., for example, observed QTc interval shortening during ASM withdrawal when patients were awake but not during sleep (Stefani et al., 2013). Akalin et al. found no effects of sodium-channel blocking ASM intake on QTc dispersion, Fialho et al. reported those on ASM polytherapy had lower cardiovascular fitness overall (Fialho et al., 2017; Akalin et al., 2003). Feldman et al. note that although clinical data does not suggest ASMs pose a significant threat in altering QTc intervals, it is also noted that there is not sufficient data to suggest there is no risk at all (Feldman and Gidal, 2013). PR interval changes have also been associated with ASM intake, specifically Lacosamide. Studies cite the possible risk of Lacosamide in increasing PR intervals, especially for patients with known cardiac conduction abnormalities (Luk et al., 2012). In studies where PR intervals were analyzed, the authors also acknowledge the possible effects various medications may have on ECG parameter results. For example, lacosamide at high doses and with rapid infusion may prolong PR intervals. However, type 1 AV-block is reported to be rare, usually insignificant and caused by minor increases in patients with borderline PR intervals at baseline (Akalin et al., 2003; Krishnan and Krishnamurthy, 2013; Ergul et al., 2013; de Sousa et al., 2017).

Also important to acknowledge are the low percentage of ECG changes found in various studies reported. In particular, it is necessary to consider how these relatively low occurrence

numbers differ from general population to understand the relevance of these results. Lamberts et al. address this question by comparing their observation to a population of 180 controls along with the PWE cohort. In this comparison, only 5% of PWE had general ECG abnormalities that were not found in the control group. While this suggests that ECG abnormalities are also found in the general population, their occurrence is still increased in PWE (Lamberts et al., 2015a). However, the low occurrence numbers highlight the importance to carefully report the differences between PWE and the general population also in other studies.

The increasing availability of long-term ambulatory ECG monitoring systems could potentially support such epilepsy diagnostics. Cardiac monitoring tools, such as implantable monitors and ECG patches, have the capability to record and analyze the ECG over extended periods of time and in the ambulatory setting. These tools make monitoring over long periods of time, as required for epilepsy seizure detection and forecasting, more feasible. Equipped with computational algorithms that detect ECG changes, potentially in a way that is personalized for the individual patient, these monitoring tools designed originally for cardiology purposes could potentially be turned into useful epilepsy diagnostics and allow a fundamentally new approach to monitor and optimize titration of therapies and to provide warning of impending seizures in epilepsy patients.

Utilization of ECG markers beyond heart rate may also help to improve responsive stimulation systems. One example may be responsive VNS targets stimulation to periods of high seizure risk that currently uses heart rate increases to trigger VNS, motivated by research that shows heart rate typically increases at seizure onset (Ryvlin et al., 2014). Alternative markers that could supplement seizure monitoring may include clinical history and seizure

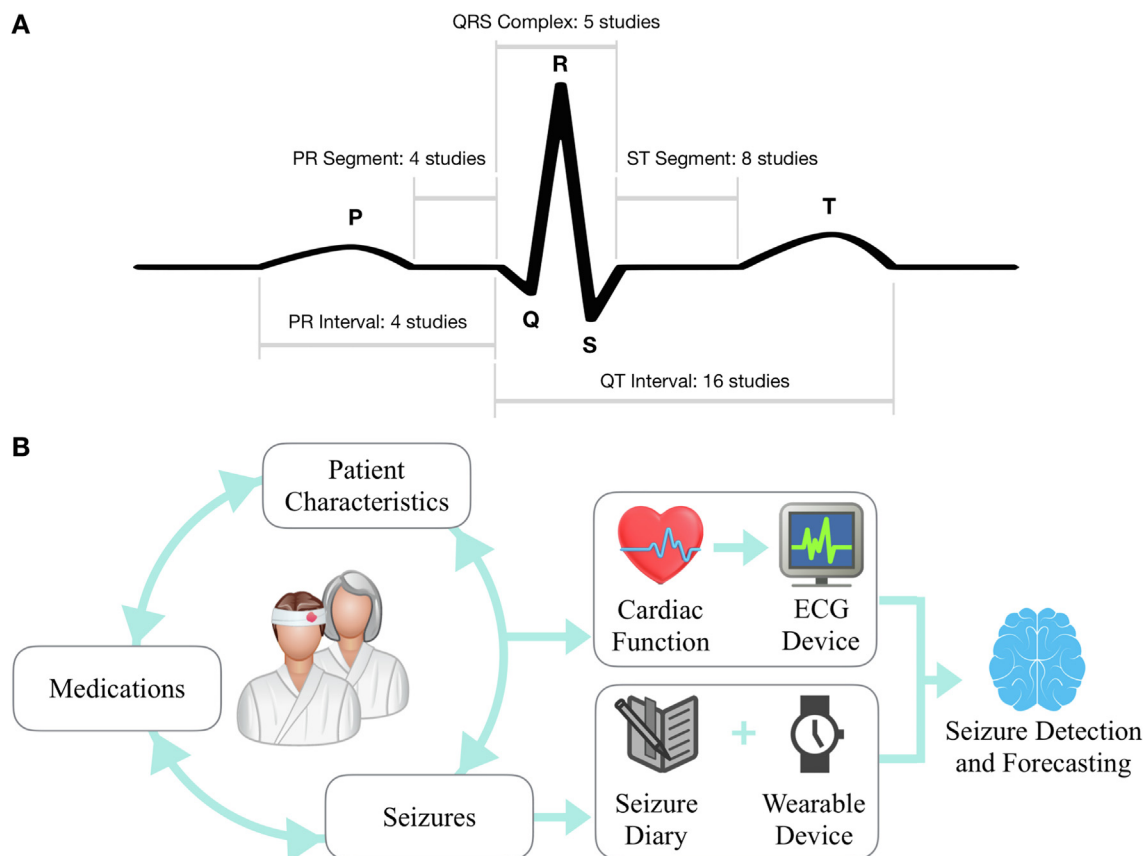


Fig. 2. A, Summary of ECG findings by number of studies. B, Mutual interactions between seizures, medications, patient specific factors and more shape ECG properties. In combination with other multi-modal monitoring devices these ECG properties may facilitate seizure detection and forecasting methods.

patterns, EEG and ECoG based biomarkers, actigraphy, EDA, or respiratory, video and sound based markers, among others. Other output parameters may include cortical or deep brain stimulation, among others (Cook et al., 2013). Utilization of further ECG changes, potentially in combination with other biomarkers (Fig. 2), to gauge seizure risk may lead to improved outcomes and quality of life of epilepsy patients by making the stimulation more sensitive and specific. Therefore, ECG should ideally be considered as one out of many modalities for seizure monitoring. It is possible that some but not all patients may benefit from it and that individual sensor combinations, including ECG, may be considered for optimal performance. ECG may be combined synergistically to other monitoring tools and modalities, potentially strengthening combined information. One may envision a set of sensors that can be combined based on patient needs or EMU assessment, utilizing the most helpful sensors and signals for each patient. Additionally, future work could benefit from using machine learning to identify the best and most efficient methods for combining multi-modal data. Algorithm based machine learning could help develop the most effective methods of seizure detection and seizure forecasting combining a variety of information and patient specific characteristics. This may create devices that are specifically attuned to individual patient multi-modal features. (Greene et al., 2007; Nasehi and Pourghassem, 2012; De Cooman et al., 2017; Meisel and Bailey, 2019; Verrier et al., 2016).

Our results need to be interpreted in the context of methodological considerations. First, there is a possibility that some studies were missed because they did not fit with our inclusion and exclusion criteria. We utilized a variety of terms in our search strategy to help screen for related manuscripts, however, it is possible that by using so many terms we unintentionally narrowed our search even further. Second, our initial title and abstract screening process may have excluded papers that discussed information we were interested in the full text but not in the abstract. Similarly, although most papers had a consensus about ECG changes around the time of a seizure, some studies showed conflicting changes. Some studies looked at specific epilepsy syndrome or seizure type and their results may not be generalized to all patients.

In summary, our literature search indicates that there are pronounced changes in the ECG in the peri-ictal state and between PWE and control groups that go beyond heart rate changes. Changes related to cardiac repolarization appear to be most prominently reported and are seen at baseline, before, during and immediately after seizures. Given that seizures are known to induce ECG changes, it is possible that such ECG alterations can be used to determine autonomic changes occurring prior to seizure onset or during seizures. The pathophysiology underlying these effects appears to be still incompletely understood. As discussed, it is important to consider variables such as differences between generalized and focal seizures (Table 1) and effects of ASM. Still, with more research, the use of ECG to detect and potentially forecast seizure onset in selected patients might help to add less-cumbersome and less-stigmatizing diagnostic tools for epilepsy management, and ECG might serve as an additional sensing modality in seizure patients, adding independent information on seizure risk and risk from seizures.

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Disclosure

Christian Meisel is part of patent applications to detect and predict clinical outcomes, and to manage, diagnose, and treat neurological conditions.

Tobias Loddenkemper serves on the Council of the American Clinical Neurophysiology Society, on the American Board of Clinical Neurophysiology, as founder and consortium PI of the pediatric status epilepticus research group (pSERG), as an Associate Editor for Wyllie's Treatment of Epilepsy 6th edition and 7th editions, and as a member of the NORSE Institute, PACS1 Foundation, and CCEMRC. He served as Associate Editor of Seizure, and served on the Laboratory Accreditation Board for Long Term (Epilepsy and Intensive Care Unit) Monitoring in the past. He is part of patent applications to detect and predict clinical outcomes, and to manage, diagnose, and treat neurological conditions, epilepsy, and seizures. Dr. Loddenkemper is co-inventor of the TriVox Health technology, and Dr. Loddenkemper, and Boston Children's Hospital might receive financial benefits from this technology in the form of compensation in the future. He received research support from the Epilepsy Research Fund, NIH, the Epilepsy Foundation of America, the Epilepsy Therapy Project, the Pediatric Epilepsy Research Foundation, and received research grants from Lundbeck, Eisai, Upsher-Smith, Mallinckrodt, Sunovion, Sage, Empatica, and Pfizer, including past device donations from various companies, including Empatica, SmartWatch, and Neuro-electrics. In the past, he served as a consultant for Zogenix, Upsher Smith, Eisai, Amzell, Engage, Elsevier, UCB, Grand Rounds, Advance Medical, and Sunovion.

He performs video electroencephalogram long-term and ICU monitoring, electroencephalograms, and other electrophysiological studies at Boston Children's Hospital and affiliated hospitals and bills for these procedures and he evaluates pediatric neurology patients and bills for clinical care. He has received speaker honorariums from national societies including the AAN, AES and ACNS, and for grand rounds at various academic centers. His wife, Dr. Karen Stannard, is a pediatric neurologist and she performs video electroencephalogram long-term and ICU monitoring, electroencephalograms, and other electrophysiological studies and bills for these procedures and she evaluates pediatric neurology patients and bills for clinical care.

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